

Alex Ning

Machine Learner

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Education

PhD in Computer Science

Sep 2026 - May 2031

New York University

Advised by [Greg Durrett](#)

BA in Mathematics and Computer Science

(2.5 year graduation) Aug 2022 - Dec 2024

University of Virginia

GPA: **3.99** / 4.0

Echols Scholar

Experience

Independent Researcher

Dec 2024 - May 2026

Collaborating with and receiving support from the **Gomes Group @ CMU** and **LIVE Robotics Lab @ UVA**

Researcher, UVA Physics Department

May 2025 - Feb 2026

NSF Center for Computer Assisted Synthesis SURF, Carnegie Mellon University

May 2024 - Aug 2024

SURF: Summer Undergraduate Research Fellowship

Performed research at the **Gomes Group @ CMU** on ML knowledge transfer methods

Works

[Visualizing LLM Latent Space Geometry Through Dimensionality Reduction](#)

Alex Ning, Vainateya Rangaraju, and Yen-Ling Kuo

ICLR Blogposts Track 2026

Blogpost: <https://iclr-blogposts.github.io/2026/blog/2026/vis-llm-latent-geometry/>

GitHub: https://github.com/Vainateya/Feature_Geometry_Visualization

[Recurrent convolutional neural networks for non-adiabatic dynamics of quantum-classical systems](#)

Alex Ning, Lingyu Yang, and Gia-Wei Chern

Physical Review E (2026)

Paper: <https://doi.org/10.1103/7dr5-68ky>

arXiv: <http://arxiv.org/abs/2412.06631>

GitHub: <https://github.com/apning/holstein-parc>

[Learning When to Stop: Adaptive Latent Reasoning via Reinforcement Learning](#)

Alex Ning, Yen-Ling Kuo, and Gabe Gomes

Preprint (2025)

arXiv: <https://arxiv.org/abs/2511.21581>

GitHub: <https://github.com/apning/adaptive-latent-reasoning>

Change-of-Basis Pruning via Rotational Invariance

Alex Ning and Vainateya Rangaraju

Preprint (2025)

arXiv: [\[2511.16061\]](https://arxiv.org/abs/2511.16061) Change-of-Basis Pruning via Rotational Invariance

GitHub: <https://github.com/apning/change-of-basis-pruning>

Presentations

Liquid Glass - Visualizing Transformer Latent Space Dynamics

Alex Ning, Vainateya Rangaraju, and Yen-Ling Kuo

UVA LLM Workshop, Charlottesville, VA (2024)

Poster: <http://links.alexning.com/f2024-lg-poster>

Quantum Density Matrix Evolution through Novel Physics-Informed Neural Networks

Alex Ning and Gia-Wei Chern

UVA Echols Research Symposium - Oral Presentation (2024)

UVA URN Research Symposium - Poster Presentation (2024)

Poster: <http://links.alexning.com/s2024-pinn-poster>

Leadership

ML @ UVA, President & Co-Founder

Aug 2023 - Dec 2024

- <https://mlatuva.org/>
- Grew organization to become the **largest** ML club at UVA with **100+** members
- Acquired and partnered with 5+ corporate and academic partners (incl. LMI, Adobe, and JHUAPL)
- Provide **mentorship** to 10+ projects, **teach ML lectures**, lead weekly **reading group**, organize **events**

Teaching

Teaching Assistant – Computer Systems & Organization 2, UVA CS Department

Jan 2024 - Dec 2024

Office hours, labs, grading

ML@UVA Lecture Series – Intro to Transformers from a Mechanistic Interpretability Perspective

Nov 2024

(1) <https://youtu.be/15tXDlIg7gGM>, (2) https://youtu.be/p_hMsXVcuDA, (3) https://youtu.be/_k_8AJSNmk0

ML@UVA Weekly Reading Group Lead

Sep 2024 - Dec 2024

<https://www.mlatuva.org/education>

ML@UVA Interest Projects Mentor

Sep 2023 - Dec 2024

Weekly mentorship to ML project teams

Languages

English (Native) | Python (Native) | PyTorch (Native) | Assembly (Unintelligible)